Introduction to Fire Dynamics

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Sciences du Risque et de la Donnée

– ENSAR –



Content

1. Background
1.1 Types of fires
1.2 Physical processes in fires
1.3 Diffusion flames vs premixed flames

2. Diffusion flames
2.1 Flame structure
2.2 Stoichiometry
2.3 Passive scalar and mixture fraction
2.4 Burke Schumann flame structure

3. Turbulent flows
3.1 Introduction
3.2 General properties

Facade fires



Many "types" of fires

Wildfires



Compartment fires

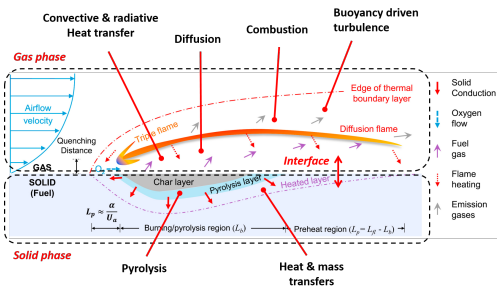


Pool fires



Jet fires









Combustion of porous media (wood) :

- ✓ **Solid phase** (heat conduction, pyrolysis, char oxidation)
- ✓ **Gas phase** (turbulent flow, heat transfer, diffusion flame)
- ✓ **Interface** (heat transfer, mass transfer)

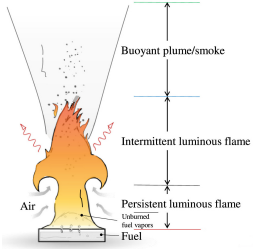






Fig 1 : Candle flame

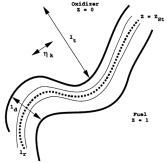
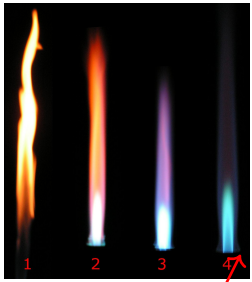


Fig 2 : Reaction zone in a diffusion flame



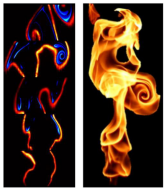


Fig 3 : Reaction zone (OH PLIF)

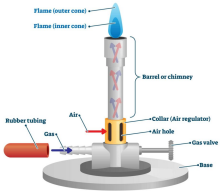


Fig 4 : Bunsen burner



Fig 1 : Candle flame

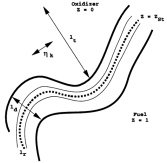
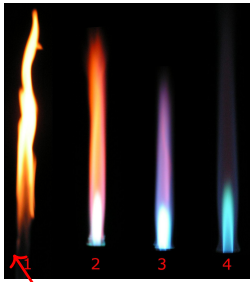


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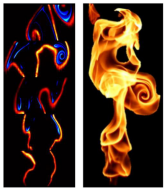


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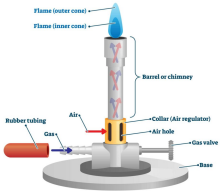


Fig 4 : Bunsen burner

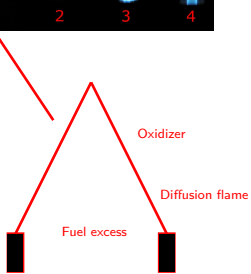




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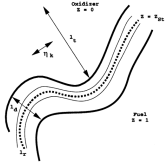


Fig 2 : Reaction zone in a diffusion flame



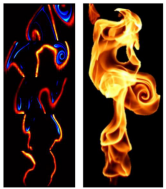


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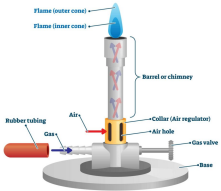
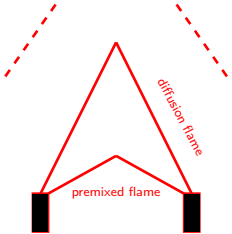


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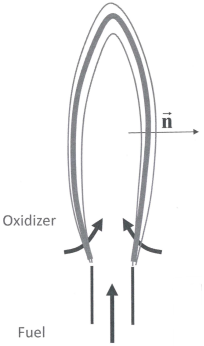


Fig 5 : Diffusion flame

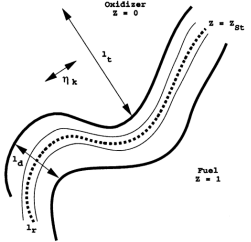


Fig 2 : Reaction zone in a diffusion flame

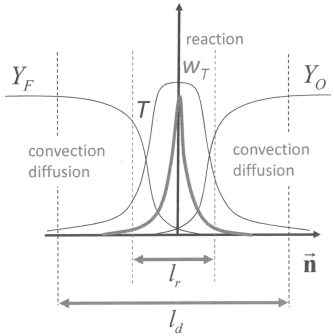


Fig 6 : Diffusion flame length scales

- ✓ l_r : Reaction length scale
- ✓ l_d : Diffusion length scale
- ✓ Da : Damköhler number
- ✓ ν_O, ν_F : stoichiometric coefficients of oxidizer and fuel

$$\frac{l_r}{l_d} \propto \frac{1}{Da^{1/(1+\nu_O+\nu_F)}}$$

Limit case : infinitely fast chemistry

✓ The mixing length scale remains finite

$$\frac{l_r}{l_d} \propto \frac{1}{Da^{1/(1+\nu_O+\nu_F)}}$$

✓ The reactive length scale becomes infinitely small :

The flame can be thought of as an interface separating combustion products + fuel excess from combustion products + oxidizer excess

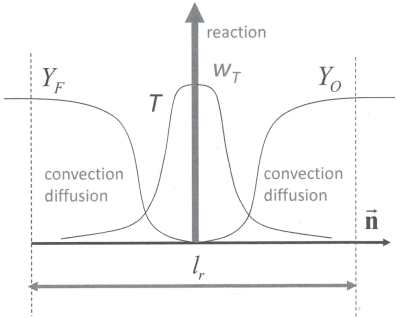


Fig 7 : Diffusion flame (infinitely fast chemistry)

$$l_d \propto \sqrt{\frac{D}{N_\xi}}$$

Stoichiometry in premixed flames

✓ **Chemical reaction**

$$\boxed{\nu'_F F + \nu'_O O \rightarrow \nu''_P P}$$

✓ **Mass stoichiometric ratio**

$$s = \left(\frac{\dot{m}_O}{\dot{m}_F}\right)_{st} = \left(\frac{\nu'_O W_O}{\nu'_F W_F}\right)_{st} = \left(\frac{Y_O}{Y_F}\right)_{st}$$

✓ **Equivalence ratio (of a given mixture)**

$$\phi = \frac{Y_F/Y_O}{(Y_F/Y_O)_{st}} = s\frac{Y_F}{Y_O}$$

$$\phi \in [0, +\infty[$$

$\dot{m}_O$ and $\dot{m}_F$ are respectively the mass flow rates of oxidizer and fuel

Equivalence ratio is a central parameter for premixed gases :

- ✓ $\phi = 1$: stoichiometric combustion
- ✓ $\phi > 1$: rich regime (fuel in excess)
- ✓ $\phi < 1$: lean regime (oxidizer in excess)

In air :

$$\boxed{Y_F = \frac{1}{1 + \frac{s}{\phi}\left(1 + 3.76\frac{W_{N_2}}{W_{O_2}}\right)}}$$

Stoichiometry in diffusion flames

✓ **Chemical reaction**

$$\boxed{\nu'_F F + \nu'_O O \rightarrow \nu''_P P}$$

✓ **Mass stoichiometric ratio**

$$s = \left(\frac{\dot{m}_O}{\dot{m}_F}\right)_{st} = \left(\frac{\nu'_O W_O}{\nu'_F W_F}\right)_{st} = \left(\frac{Y_O}{Y_F}\right)_{st}$$

✓ **Equivalence ratio**

$$\phi = s\frac{Y^{\infty}_{F,fuel}}{Y^{\infty}_{O,ox}}$$

⇒ First definition of equivalence ratio

⇒ Characterizes the local structure of the flame when 2 streams (fuel and oxidizer) interact

⇒ It does not represent the overall behavior of the process

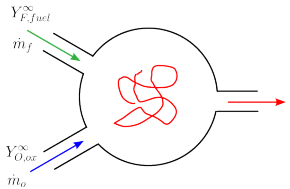


Fig 8 : Perfectly stirred reactor (PSR)

⇒ Global equivalence ratio ϕ_g is introduced

$$\phi_g = s\frac{\dot{m}_F}{\dot{m}_O}$$

$\dot{m}_i$: flow rate of a species i in fuel or oxidizer inlet

$$Y_i = \frac{m_i}{m_{tot}} \qquad \dot{m}_i = Y_i \dot{m}_{tot}$$

$$\phi_g = s\frac{Y^{\infty}_{F,fuel}\dot{m}_{F,tot}}{Y^{\infty}_{O,ox}\dot{m}_{O,tot}}$$

$$\phi_g = \phi \frac{\dot{m}_{F,tot}}{\dot{m}_{O,tot}}$$

✓ For premixed conditions, fuel and oxidizer are carried by the same stream
$\Rightarrow \dot{m}_{F,tot} = \dot{m}_{O,tot}$
$\Rightarrow \phi_g = \phi$

✓ What is the representativeness of ϕ for ***separated*** injection of fuel and oxidizer ?

In the following, we do not consider partially premixed conditions :

$$Y^{\infty}_{F,fuel} = Y^{\infty}_{F} \qquad Y^{\infty}_{O,ox} = Y^{\infty}_{O}$$

✓ **Global reaction**

$$\boxed{\nu'_F F + \nu'_O O \rightarrow \nu''_P P}$$

✓ Balance equation for **mass fraction** Y_k of each species (F, O, P):

$$\boxed{\underbrace{\partial_t(\rho Y_k)}_{storage} + \underbrace{\nabla\cdot(\rho u Y_k)}_{convection} = \underbrace{\nabla\cdot(\rho D \nabla Y_k)}_{diffusion} + \underbrace{\dot{\omega}_k}_{production}}$$

✓ Species reaction rates $\dot{\omega}_k$ are all related to the single-step reaction rate $\mathcal{Q}$:

$$\boxed{\dot{\omega}_k = W_k \nu_k \mathcal{Q}, \qquad \nu_k = \nu''_k - \nu'_k}$$

✓ The oxidizer reaction rate is linked to the fuel reaction rate :

$$\boxed{\dot{\omega}_O = s\dot{\omega}_F}$$

✓ Conservation equation for **energy** (temperature) :

$$\partial_t (\rho Cp T) + \nabla \cdot (\rho Cpu T) = \nabla \cdot (\lambda \nabla T) + \dot{\omega}_T$$

$\dot{\omega}_T = -Q\dot{\omega}_F$ (Q : heat of reaction per unit mass of fuel)

Using these relations, and assuming $D_i = D$, the conservation equations for fuel, oxidizer and temperature become :

① $\partial_t (\rho Y_F) + \nabla \cdot (\rho u Y_F) = \nabla \cdot (\rho D \nabla Y_F) + \dot{\omega}_F$

② $\partial_t (\rho Y_O) + \nabla \cdot (\rho u Y_O) = \nabla \cdot (\rho D \nabla Y_O) + s\dot{\omega}_F$

③ $\partial_t (\rho T) + \nabla \cdot (\rho u T) = \nabla \cdot (\rho D \nabla T) - \dfrac{Q}{Cp}\dot{\omega}_F$

$$Le_i = \frac{\alpha}{D_i} = \frac{\text{rate of heat diffusion}}{\text{rate of chemical species diffusion}}$$

$$\alpha = \frac{\lambda}{\rho Cp}$$

$$Le = \frac{\lambda}{\rho Cp D}$$

$$Le = 1 \Rightarrow D_i = \alpha \Rightarrow \lambda = \rho Cp D$$

Diffusion flames – Passive scalar and mixture fraction –

✓ **Problem**
The presence of non-linear source terms makes direct solution difficult (especially in multi-reacting flows)

✓ **Shvab-Zel'dovich idea**
$\Rightarrow$ construct linear combinations of species mass fractions and enthalpy such that the chemical source terms cancel
$\Rightarrow$ The resulting transformed variables (conserved scalars) obey pure convection-diffusion equations (without chemical sources)

✓ **Shvab-Zel'dovich transformation**
Mathematical technique that aims to simplify the coupled reactive transport equations of chemical species and energy by constructing linear combinations of conserved scalars in which chemical source terms cancel out

✓ **Outcome**
$\Rightarrow$ The complex reactive problem is reduced to solving one or more passive scalar transport equations
$\Rightarrow$ All other quantities (T, Y_k) can then be reconstructed a posteriori from the conserved scalars and the stoichiometry of the reactions

✓ **Assumptions**
$\Rightarrow$ Assume constant Lewis number ($Le = 1$) to make the transformation exact
$\Rightarrow$ Single-step global reactions are typically considered for analytical tractability

✓ **Shvab-Zel'dovich variables** (by combining eq. 1 – 3)

$$\boxed{Z_1 = sY_F - Y_O}$$

$$\boxed{Z_2 = Y_F + \frac{Cp}{Q}T}$$

$$\boxed{Z_3 = Y_O + s\frac{Cp}{Q}T}$$

✓ All linear combinations lead to the same balance equation

$$\boxed{\partial_t(\rho Z) + \nabla \cdot (\rho u Z) = \nabla \cdot (\rho D \nabla Z)}$$

The 3 variables Z_1, Z_2, Z_3 follow the same balance eq. but have different boundary conditions summarized below :

passive scalar Z	fuel value Z_i^F	oxidizer value Z_i^O
Z_1	sY_F^∞	$-Y_O^\infty$
Z_2	$Y_F^\infty + \frac{Cp}{Q}T_F^\infty$	$\frac{Cp}{Q}T_O^\infty$
Z_3	$s\frac{Cp}{Q}T_F^\infty$	$Y_O^\infty + s\frac{Cp}{Q}T_O^\infty$

Table 1: Boundary conditions of Z_i

Normalized variables ξ_j are defined :

$$\boxed{\xi_j = \frac{Z_j - Z_j^O}{Z_j^F - Z_j^O}}$$

All reduced variables ξ_j follow the same convection / diffusion balance equation :

$$\boxed{\partial_t(\rho\xi) + \nabla \cdot (\rho u \xi) = \nabla \cdot (\rho D \nabla \xi)}$$

and have the same boundary conditions : $\xi_j = 1$ in the fuel stream and $\xi_j = 0$ in the oxidizer stream

$$\boxed{\xi_1 = \xi_2 = \xi_3 = \xi}$$

✓ ξ is the **mixture fraction**

$\Rightarrow$ measures the local fuel / oxidizer ratio

Expressing ξ using boundary conditions leads to :

$$\boxed{\xi = \frac{sY_F - Y_O + Y_O^\infty}{sY_F^\infty + Y_O^\infty} = \frac{\frac{Cp}{Q}(T - T_O^\infty) + Y_F}{\frac{Cp}{Q}(T_F^\infty - T_O^\infty) + Y_F^\infty} = \frac{s\frac{Cp}{Q}(T - T_O^\infty) + Y_O - Y_O^\infty}{s\frac{Cp}{Q}(T_F^\infty - T_O^\infty) - Y_O^\infty}}$$

The first part of ξ may be recast as :

$$\boxed{\xi = \frac{1}{\phi + 1}\left(\phi\frac{Y_F}{Y_F^\infty} - \frac{Y_O}{Y_O^\infty} + 1\right)}$$

ϕ is the equivalence ratio :

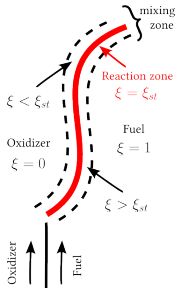
$$\boxed{\phi = s\frac{Y_F^\infty}{Y_O^\infty}}$$

At stoichiometry $\Rightarrow$ complete combustion (no excess of oxidizer or fuel)

$$\boxed{\xi_{st} = \frac{1}{\phi + 1}}$$

Assumptions :

- ✓ Infinitely fast chemistry
  - ⇒ The flame front is considered infinitely thin and located at $\xi = \xi_{st}$
  - ⇒ Fuel and oxidizer are consumed instantaneously when they meet at the stoichiometric ratio
  - ⇒ On both sides of the flame, temperature and species profiles are linear versus ξ (slopes are discontinuous at the flame front)
  - ⇒ The reaction rate is infinite at the flame front and zero elsewhere.
- ✓ One-step global reaction
- ✓ Unity Lewis number ($Le = 1$)
- ✓ Constant density



✓ Solution for **species mass fractions** in the Burke-Schumann flame structure :

✓ **Lean region** : $\xi < \xi_{st}$

$$\boxed{\begin{aligned} &Y_F = 0, \\ &Y_O(\xi) = Y_O^{\infty}\left(1 - \frac{\xi}{\xi_{st}}\right) \end{aligned}}$$

The remaining oxidizer is proportional (linear dependence) to the deficit in ξ compared to the stoichiometric value ξ_{st}.

✓ **Rich region** : $\xi > \xi_{st}$

$$\boxed{\begin{aligned} &Y_O = 0, \\ &Y_F(\xi) = Y_F^{\infty}\left(\frac{\xi - \xi_{st}}{1 - \xi_{st}}\right) \end{aligned}}$$

The remaining fuel is proportional (linear dependence) to the excess in ξ compared to the stoichiometric value ξ_{st}.

✓ Solution for **Temperature** in the Burke-Schumann flame structure :

$$T_{st} = \xi_{st} T_F^{\infty} + (1 - \xi_{st}) T_O^{\infty} + \xi_{st} \frac{Q Y_F^{\infty}}{Cp}$$

✓ Outside the flame sheet there is no chemical source

⇒ Under the assumptions $Le = 1$ (or equal diffusivities) and constant Cp, the steady transport in ξ-space leads to a linear relation for conserved scalars

⇒ Consequently the temperature is linear in ξ on each side of the sheet

✓ With boundary conditions $T(\xi = 0) = T_O^{\infty}$ and $T(\xi = 1) = T_F^{\infty}$ and the jump condition $T(\xi_{st}) = T_{st}$, we get:

✓ For oxidizer side $0 \leq \xi \leq \xi_{st}$

$$\boxed{T(\xi) = T_O^\infty + (T_{st} - T_O^\infty)\,\frac{\xi}{\xi_{st}}}$$

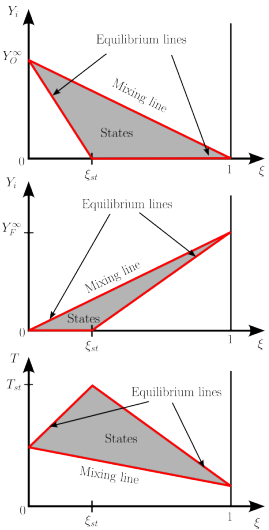
✓ For fuel side $\xi_{st} \leq \xi \leq 1$

$$\boxed{T(\xi) = T_F^\infty + (T_{st} - T_F^\infty)\,\frac{1-\xi}{1-\xi_{st}}}$$

✓ These 2 linear segments meet at the stoichiometric point (ξ_{st}, T_{st}).

✓ Check limits :

- ✓ $\xi = 0 \Rightarrow T = T_O^\infty$
- ✓ $\xi = 1 \Rightarrow T = T_F^\infty$
- ✓ $\xi = \xi_{st} \Rightarrow T = T_{st}$



✓ **Oxidizer** Y_O

⇒ Linear decrease of Y_O up to ξ_{st} reflects the diffusive mixing between fuel jet and surrounding air

⇒ Beyond ξ_{st}, $Y_O = 0$: oxygen is immediately consumed upon contact with fuel → no oxidizer penetration into the fuel zone

⇒ Flame acts as a surface impermeable to the oxidizer

✓ **Fuel** Y_F

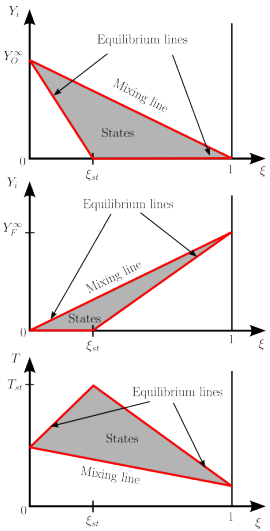
⇒ Same mirror-like behavior as $Y_O(\xi)$: linear decrease to 0 at the stoichiometric point

⇒ For $\xi < \xi_{st}$, $Y_F = 0$: fuel is consumed at the flame sheet → no fuel penetration into the oxidizer zone

✓ **Temperature** $T(\xi)$

⇒ Maximum temperature is located at $\xi = \xi_{st}$

⇒ On the air side, not enough fuel → little reaction; on the fuel side, not enough oxygen → little reaction; at ξ_{st}, perfect balance → maximum energy release, The flame is therefore geometrically located on the surface $\xi = \xi_{st}$



✓ **Idealized nature of the Burke-Schumann flame**

⇒ In this model, the flame has no thickness → it is a surface-type discontinuity (like a reactive membrane)

⇒ All the reaction is confined within an infinitely thin layer → no preheated zone or internal structure

⇒ Asymptotic approximation valid for $Da \to \infty$ (large Damköhler number : infinitely fast chemistry)

✓ **Physical implications**

⇒ Purely diffusion-controlled regime : Neither aerodynamics, turbulence, nor kinetics determine the flame location → only stoichiometry does

⇒ Maximum temperature: Since it occurs exactly at ξ_{st} the value of T_{st} directly depends on:

✓ the heat of reaction of the fuel

✓ the average heat capacity of the mixture

✓ stoichiometric ratio (indirectly on the nature of the fuel)

✓ **What is flow in fluid mechanics ?**

✓ A flow refers to the movement of a fluid (liquid or gas) in space and time, described by variables such as **velocity**, **pressure**, **density**, and **temperature**

✓ How this motion is organized depends primarily on the balance between **viscous forces** (which smooth out motion) and **inertial forces** (which promote irregularity and instability)

✓ **Laminar Flow**

✓ **Definition:**

⇒ A laminar flow is one in which fluid particles move in smooth, orderly layers (laminae) that slide past each other without mixing

✓ **Characteristics:**

⇒ The velocity at a given point is nearly constant in time

⇒ Neighboring fluid layers do not mix significantly

⇒ The flow is predictable and stable

⇒ Momentum, heat, and mass are transported mainly by molecular diffusion (no eddies)



Fig x : Flows at the Noisiel dam

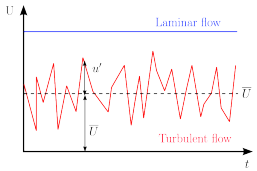


Fig x : Flows at the Noisiel dam

✓ **Turbulent Flow**

✓ **Definition:**

⇒ A turbulent flow is a flow regime characterized by irregular, chaotic motion where velocity and pressure fluctuate strongly in time and space



✓ **Characteristics:**

⇒ The flow contains a wide range of eddies (vortices) of different sizes

⇒ Fluid elements move in complex, random paths

⇒ There is strong mixing of momentum, heat, and mass — much greater than by molecular diffusion alone

⇒ The mean flow can still be steady, but instantaneous velocities are highly unsteady

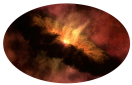
✓ **Transition from laminar to turbulent flow**

The transition depends on a dimensionless number called the Reynolds number:

$$Re = \frac{\rho U L}{\mu} = \frac{U L}{\nu}$$

ρ: fluid density ($\mathrm{kg.m^{-3}}$)
U: characteristic velocity ($\mathrm{m.s^{-1}}$)
L: characteristic length (m)
μ: dynamic viscosity ($\mathrm{kg.m^{-1} \cdot s^{-1}}$)
ν: kinematic viscosity ($\mathrm{m^2.s^{-1}}$)







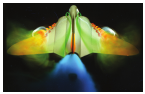


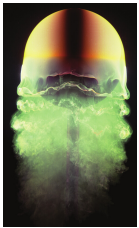
Table 2: Typical thresholds between laminar, transitional, and turbulent regimes.

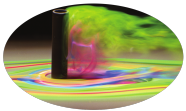
Flow type	Reynolds number (approx.)	Example
Laminar	$Re < 2300$	Flow in a small, slow pipe
Transitional	$2300 < Re < 4000$	Flow starting to fluctuate
Turbulent	$Re > 4000$	Fast flow in a large pipe

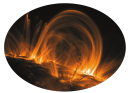


Table 3: Physical differences between laminar and turbulent flows.

Property	Laminar flow	Turbulent flow
Velocity field	Smooth, steady	Fluctuating, chaotic
Mixing	Weak	Strong
Energy dissipation	Low	High
Predictability	High	Low (statistical treatment needed)
Dominant mechanism	Molecular viscosity	Eddy viscosity (turbulent transport)







In fluid mechanics, turbulence does not appear spontaneously – it emerges through a sequence of instabilities that progressively disrupt a laminar flow
⇒ Understanding this process is central to explaining why some flows remain smooth while others become chaotic

1. **Laminar flow : the stable state**

✓ At low Reynolds numbers, viscous forces dominate over inertia
✓ In this regime, fluid particles move in smooth, parallel layers, and any disturbance is rapidly damped by viscosity.

2. **The role of instabilities**

✓ As the Reynolds number increases, the balance between viscous and inertial forces changes
✓ Above a critical value Re_c, the laminar flow becomes unstable – small perturbations no longer decay but amplify over time
✓ Physically, this means that the fluid has enough inertial energy to feed perturbations faster than viscosity can damp them
⇒ The result is the appearance of organized oscillations or vortices, which mark the first stage of transition

3 From instability to transition

✓ The laminar – turbulent transition rarely happens abruptly

✓ Instead, it proceeds through a cascade of instability mechanisms, each introducing more spatial and temporal complexity

✓ Typical examples include:

- ✓ Kelvin–Helmholtz vortices in shear layers or wakes: roll-up of velocity gradients.
- ✓ von Kármán vortex streets: periodic shedding of vortices behind bluff bodies.
- ✓ Rayleigh-Taylor instabilities in stratified fluids: interpenetration of layers with different densities.

⇒ These structures grow, interact, and eventually break down into disordered motions – the onset of turbulence.

4 Nonlinear saturation and breakdown

✓ When perturbations become large enough, nonlinear effects can no longer be neglected

✓ Different instability modes interact, exchanging energy and distorting the mean flow

✓ This stage is characterized by:

- ✓ Secondary instabilities (three-dimensional deformation of initial vortices)
- ✓ Rapid energy transfer across scales
- ✓ The formation of localized turbulent spots or patches

⇒ Once these turbulent regions merge and occupy the whole flow, the system reaches a fully developed turbulent state

5. **Fully developed turbulence**

✓ In this regime:

- ✓ The instantaneous velocity is highly irregular
- ✓ Energy cascades from large eddies (where it is injected) to small eddies (where it is dissipated)
- ✓ The flow must be described statistically (mean flow, fluctuations, spectra, correlations)

✓ The transition therefore represents a loss of spatial and temporal coherence, but not a loss of physical structure — turbulence still follows well-defined laws of energy transfer and dissipation (e.g., Kolmogorov 1941 theory).

Table 4: Summary of the conceptual differences between laminar and turbulent flows.

Aspect	**Laminar flow**	**Turbulent flow**
Nature of motion	Ordered, smooth	Disordered, chaotic
Predictability	Deterministic	Statistical (requires averaging)
Dominant scales	Single characteristic scale	Broad spectrum of scales (energy cascade)
Mixing and diffusion	Molecular diffusion dominates	Eddy diffusion dominates
Energy transport	Primarily viscous	Primarily inertial
Momentum transfer	Molecular viscosity	Turbulent (eddy) viscosity
Mathematical description	Exact solutions possible (Navier–Stokes)	Requires modeling and closure assumptions

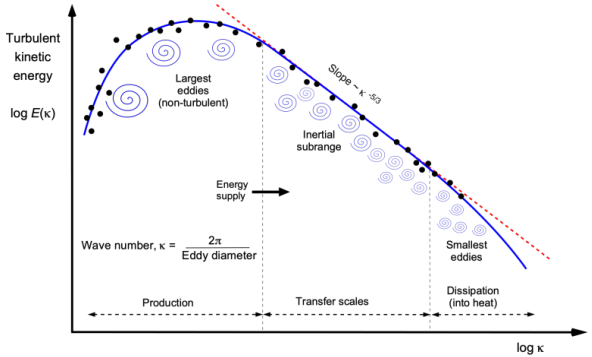


Fig x : Kolmogorov's energy cascade

✓ The transition to turbulence can thus be viewed as a competition between two processes:

- ✓ Dissipation due to viscosity, which smooths the flow,
- ✓ Production of perturbation energy by inertial mechanisms (shear, curvature, buoyancy).

✓ When energy production exceeds dissipation, the laminar state can no longer be maintained — the flow reorganizes into a turbulent one.

1 Kelvin–Helmholtz instability



The Kelvin–Helmholtz instability occurs whenever two fluid layers move at different velocities along a shared interface

- ✓ Upper layer (U_1), Lower layer (U_2), $U_1 > U_2$
- ✓ At $y = 0$, there is an interface between them – initially flat
- ✓ A small bump (perturbation) appears on the interface, like a tiny wave crest

Because of the bump :

- ✓ The streamlines of the faster fluid (upper layer) are deflected upward over the crest
- ✓ The streamlines compress (come closer together) above the bump → meaning the local velocity increases there
- ✓ Below the bump, streamlines of the slower layer spread apart → velocity decreases locally

This comes directly from the continuity of the flow – when streamlines converge, the local velocity magnitude must increase to maintain the same volume flux.

✓ For a steady, inviscid, incompressible flow along a streamline, Bernoulli's equation states:

$$p + \frac{1}{2}\rho u^2 + \rho g h = \text{constant}$$

✓ Ignoring gravity for now (flat interface approximation), we get:

$$p + \frac{1}{2}\rho u^2 = \text{constant}$$

✓ Hence, if u increases $\rightarrow$ p decreases, and vice versa

Let's focus on the pressure field near the perturbation :

Table 5: Application of Bernoulli's principle to a small bump on the interface.

Location	Local velocity	Local pressure	Effect
Above the bump (fast layer)	$u_1 > U_1$	$p_1 < p_\infty$	Lower pressure
Below the bump (slow layer)	$u_2 < U_2$	$p_2 > p_\infty$	Higher pressure

So, the pressure on the lower side of the bump is higher than on the upper side.

That pressure difference across the interface exerts a net upward force on the bump:

$$\Delta p = p_2 - p_1 > 0$$

This force amplifies the initial perturbation $\rightarrow$ the bump grows higher
This is a positive feedback mechanism: The larger the bump $\rightarrow$ the more the streamlines deflect $\rightarrow$ the stronger the pressure contrast $\rightarrow$ the faster the growth.
Thus, the flow's kinetic energy (mean shear) is converted into wave energy at the interface.

The Rayleigh–Taylor instability occurs when a heavy fluid lies above a lighter one in a gravitational field, i.e. the configuration is gravitationally unstable.
$\rho_1 > \rho_2$, with ρ_1 above ρ_2, $\mathbf{g} = -g\,\mathbf{e}_y$
If the interface between the two fluids is slightly perturbed:

- ✓ A small upward bump of light fluid appears
- ✓ The heavy fluid falls under gravity, while the lighter fluid rises into the depressions
- ✓ These motions amplify the initial disturbance rather than damp it

In other words, gravity converts potential energy into kinetic energy, driving the growth of the instability.
As the instability grows:

- ✓ Spikes correspond to regions where the heavier fluid penetrates downward into the lighter one. Because gravity acts downward, any small downward displacement of the heavy fluid amplifies — the heavy fluid "falls" further under its own weight.
- ✓ Bubbles are the opposite: the lighter fluid rises upward into the heavy one. This happens because buoyancy drives the light fluid to move upward where the heavy fluid has been displaced.

At this stage, the flow is still relatively organized — the interface has a wavy shape with crests and troughs that sharpen over time. The potential energy of the density stratification (heavy on top of light) is converted into kinetic energy of motion.

As the spikes and bubbles accelerate under gravity g, the velocity difference between adjacent regions of light and heavy fluid increases.
At their boundaries, shear layers develop — similar to the Kelvin–Helmholtz mechanism.
These shear layers generate vorticity (rotational motion) because of the large velocity gradients.
The vorticity tends to wrap around the spikes and bubbles, producing mushroom-like structures characteristic of RT instability (you often see them in astrophysical or inertial confinement plasma images).
The acceleration $a \sim g(\rho_1 - \rho_2)/(\rho_1 + \rho_2)$ drives continuous stretching of the interface. Vorticity ω grows due to baroclinic torque, given by:

$$\frac{D\omega}{Dt} = \frac{1}{\rho^2}(\nabla\rho \times \nabla p)$$

Since pressure and density gradients are not aligned near the interface, their cross product creates rotation.

Table 6: Summary of the physical stages of the Rayleigh–Taylor instability.

Stage	Physical process	Key mechanism	Outcome
Linear growth	Small perturbations on the interface leading to instability growth	Buoyancy amplifies the density inversion	Exponential amplification of disturbances
Nonlinear regime	Formation of spikes (heavy fluid descending) and bubbles (light fluid rising)	Conversion of potential to kinetic energy	Accelerated motion of coherent structures
Vorticity generation	Shear and baroclinic effects at the interface	$\nabla\rho \times \nabla p \neq 0$ induces vorticity	Development of mushroom-shaped vortices
Turbulent mixing	Interaction and merging of spikes and bubbles	Secondary Kelvin–Helmholtz instabilities	Fully developed Rayleigh–Taylor mixing zone